

LTA analysis - Descriptive Stats

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Reading in dataset from 2009

2009 dataset now includes a total of 8,702 observations (matching analytic dataset in published paper) and includes a total of 79 variables - including indexno

```
lca_pcp_2009 <- read.table(  
  file.path(todd_dir, "lca_pcp_2009_1.24.23.dat"),  
  sep = ",",  
  header = FALSE,  
  fill = TRUE,  
  comment.char = "",  
  quote = ""  
)  
  
#Changing all values of -9999 to missing  
lca_pcp_2009[lca_pcp_2009==-9999] <- NA
```

Reading in dataset from 2019

2019 dataset now includes a total of 7,929 observations (matching analytic dataset in published paper) and includes a total of 79 variables - including indexno

```
lca_pcp_2019 <- read.table(  
  file.path(todd_dir, "lca_pcp2019_1.24.23.dat"),  
  sep = ",",  
  header = FALSE,  
  fill = TRUE,  
  comment.char = "",  
  quote = ""
```

```
)

#Changing all values of -9999 to missing
lca_pcp_2019[lca_pcp_2019==-9999] <- NA

names(lca_pcp_2019) <- vars_2009
```

Creating merged dataset

First we want to rename all variables, except indexno - our ID - with subscript denoting year. Also adding an indicator variable to denote whether the indexno is in 2009 and/or in 2019.

```
lca_pcp_2009_renamed <- lca_pcp_2009 %>%
  rename_with(~ paste0(.x, "_2009"), -indexno, -in_2009) %>%
  mutate(in_2009 = 1L)

lca_pcp_2019_renamed <- lca_pcp_2019 %>%
  rename_with(~ paste0(.x, "_2019"), -indexno, -in_2019) %>%
  mutate(in_2019 = 1L)
```

Now merging dataset - including all indexno in **either** 2009 and/or 2019

```
lca_pcp_merged <- lca_pcp_2009_renamed %>%
  full_join(lca_pcp_2019_renamed, by = "indexno") %>%
  mutate(
    in_2009 = coalesce(in_2009, 0L),
    in_2019 = coalesce(in_2019, 0L)
  )
```

Now looking at descriptive statistics

```
table(lca_pcp_merged$in_2009, lca_pcp_merged$in_2019)
```

```
      0      1
0      0 3312
1 4085 4617
```

```

num_vars <- lca_pcp_merged %>%
  select(where(is.numeric)) %>%
  select(-indexno, -in_2009, -in_2019) %>%
  names()

summary_table <- lca_pcp_merged %>%
  select(indexno, in_2009, in_2019, all_of(num_vars)) %>%
  pivot_longer(
    cols = all_of(num_vars),
    names_to = "variable",
    values_to = "value"
  ) %>%
  mutate(
    year = case_when(
      str_ends(variable, "_2009") ~ "2009",
      str_ends(variable, "_2019") ~ "2019"
    ),
    base_var = str_remove(variable, "_2009|_2019")
  ) %>%
  group_by(base_var) %>%
  mutate(
    is_binary = all(na.omit(value) %in% c(0, 1))
  ) %>%
  ungroup() %>%
  mutate(
    group = case_when(
      in_2009 == 1 & in_2019 == 0 & year == "2009" ~ "2009_only",
      in_2009 == 1 & in_2019 == 1 & year == "2009" ~ "both_2009",
      in_2009 == 1 & in_2019 == 1 & year == "2019" ~ "both_2019",
      in_2009 == 0 & in_2019 == 1 & year == "2019" ~ "2019_only",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(group)) %>%
  group_by(base_var, is_binary, group) %>%
  summarise(
    mean = mean(value, na.rm = TRUE),
    sd    = sd(value, na.rm = TRUE),
    n1    = sum(value == 1, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(

```

```

value_display = if_else(
  is_binary,
  paste0(
    round(mean * 100), "% (N = ", n1, ")"
  ),
  paste0(
    round(mean, 1),
    " (",
    round(sd, 1),
    ")"
  )
)
) %>%
select(base_var, group, value_display) %>%
pivot_wider(
  names_from = group,
  values_from = value_display
) %>%
arrange(base_var)

summary_table

```

```

# A tibble: 78 x 5
  base_var   `2009_only`   `2019_only`   both_2009   both_2019
  <chr>      <chr>          <chr>        <chr>      <chr>
1 _0to5      8% (N = 318)    22% (N = 735) 6% (N = 268) 0% (N = 0)
2 _11to20    26% (N = 1075) 27% (N = 908) 39% (N = 1808) 21% (N = 971)
3 _21to30    24% (N = 980)  11% (N = 376) 28% (N = 1283) 39% (N = 1808)
4 _31more    25% (N = 1008) 10% (N = 324) 12% (N = 553) 40% (N = 1836)
5 _6to10     17% (N = 688)  29% (N = 967) 15% (N = 703) 0% (N = 0)
6 adi_state  4.8 (3)         4.4 (3)        4.6 (3)      4.5 (3)
7 age        47.9 (12.3)    40.8 (10.9)    45.3 (9.2)   55.3 (9.2)
8 age35below 19% (N = 792)  39% (N = 1302) 15% (N = 697) 0% (N = 0)
9 age36to45  27% (N = 1108) 36% (N = 1198) 39% (N = 1796) 15% (N = 697)
10 age46to55 24% (N = 981)  13% (N = 415) 32% (N = 1462) 39% (N = 1796)
# i 68 more rows

```

Formatting this table so this can be output - including column sample sizes

```

n_2009_only <- sum(lca_pcp_merged$in_2009 == 1 & lca_pcp_merged$in_2019 == 0)
n_both      <- sum(lca_pcp_merged$in_2009 == 1 & lca_pcp_merged$in_2019 == 1)

```

```

n_2019_only <- sum(lca_pcp_merged$in_2009 == 0 & lca_pcp_merged$in_2019 == 1)

summary_table_n <- summary_table

colnames(summary_table_n) <- c(
  "Variable",
  paste0("2009 only (N = ", n_2009_only, ")"),
  paste0("Both years - 2009 (N = ", n_both, ")"),
  paste0("Both years - 2019 (N = ", n_both, ")"),
  paste0("2019 only (N = ", n_2019_only, ")")
)

```

Output summary table

```

write.csv(
  summary_table_n,
  file = "lca_pcp_summary_table_041623.csv",
  row.names = FALSE
)

```